

PlantProcess IQ -- Analytics Methodology and Trust

Document class	Methodology and Trust
Audience	Process and quality engineers, executives, data reviewers
Product	PlantProcess IQ (PPIQ) -- SOU Industrial Software
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PlantProcess IQ is built to be trusted by a skeptical engineer. This document explains how its findings are produced and why the way they are framed is deliberate.

Correlation, not causation

The product performs correlation analysis. It identifies statistical relationships between process parameters and outcomes such as defects, downtime, and KPI movement, and it frames every result as a suspected contributor. It does not assert a guaranteed root cause. This is an honest description of what data analysis over plant history can and cannot establish.

Population is always stated

Every analysis reports the population it was computed on and the records it excluded -- for example, how many objects of a candidate set carried the data required for the analysis. The product refuses silent survivorship: it never states a driver without stating its population, because the records that were dropped may be exactly where a problem lives. Where the data is not yet sufficient to support an advanced finding, the product shows a readiness state and an honest countdown rather than fabricating an answer.

Method set

Relationships are assessed with an established set of statistical methods rather than a single naive measure -- rank correlation, mutual information, regularized regression, multicollinearity checks, and bootstrap stability. Using several complementary methods guards against a spurious result from any one of them.

Multiple-testing control and stratification

When many parameters are tested at once, some will appear related by chance. PlantProcess IQ controls for this with false-discovery-rate control, and it stratifies by visible confounders so that an apparent relationship is not an artifact of a third factor. Where a likely confounder cannot be measured, the product names it rather than ignoring it.

Blended provenance

Where a finished object draws from more than one upstream source -- for example, a product spanning a transition between two production units -- the product reports a weighted, shared attribution between

them rather than assigning it to a single fabricated origin.

Quantified value with an abstain path

A finding can be expressed as a bounded economic estimate -- a range, computed on the plant's own data, with every input drillable to its source, and with an explicit abstain path when the data does not support a number. This allows a finding to be weighed economically without overstating its precision.

Grounded assistance

Where a natural-language assistant is provided, deterministic engines compute and rank, and the assistant only explains the result with citations. It cannot present a number that is not grounded in a resolvable piece of evidence, and any claim it makes can be audited back to that evidence.

The honesty contract, summarized

- Results are suspected contributors and statistical patterns, not guaranteed causes.
- Every result states its method, sample size, filters, and exclusions.
- Insufficient data yields a readiness state, never a fabricated answer.
- A confident wrong answer is worse than an honest "still collecting data," and the product is built to prefer the latter.

This posture is not a limitation to be apologized for. It is the reason the product's findings can be relied upon in a plant where the cost of acting on a wrong conclusion is high.